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Evaluation of 'eco BRICK Innovio' Polymer-Based Aggregate
ASTM C331 Testing
CTLGroup Project No. 201338

Dear Shlomi:

As requested, CTLGroup has conducted ASTM C331 testing on 'eco BRICK Innovio' lightweight aggregate samples provided by you. Seven aggregate samples meeting the nominal size (U.S. Imperial) requirements of '3/8" – No.4', 'No.4 – No.8', 'No.8 – No.16', 'No.16 – No.30', 'No.30 – No.50', 'No.50 – No.100', and 'No.100 – Pan' were received by CTLGroup for testing. The scope of work included evaluation of a blend of 'as-received' aggregates and evaluation of concrete specimens fabricated using a blend of 'as-received' aggregates for various performance criteria listed in ASTM C331, *Standard Specification for Lightweight Aggregates for Concrete Masonry Units*.

It should be noted that the aggregates under evaluation don't meet any of the three general aggregate types listed in Section 4 of ASTM C331, which are listed below:

1. Aggregates prepared by expanding, pelletizing, or sintering products such as blast-furnace slag, clay, diatomite, fly ash, shale, or slate
2. Aggregates prepared by processing natural materials, such as pumice, scoria, or tuff
3. Aggregates consisting of end products of coal or coke combustion.

Because the recycled polymer-based aggregates tend to behave differently compared to the aggregate materials mentioned above, especially at high temperatures, some of the ASTM C331 performance criteria listed for Loss on Ignition (ASTM C114) and Autoclave Expansion (ASTM C151) are not applicable. Hence, upon discussion with you, these two test methods were excluded from ASTM C331 testing scope. During the testing program, the following tests were conducted (with modifications as specified in ASTM C331, if applicable):

- ASTM C136, *Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates*
- ASTM C40, *Standard Test Method for Organic Impurities in Fine Aggregates for Concrete*
- ASTM C29, *Standard Test Method for Bulk Density (Unit Weight) and Voids in Aggregate*
- ASTM C142, *Standard Test Method Clay Lumps and Friable Particles in Aggregates*
- ASTM C157, *Standard Test Method for Length Change of Hardened Hydraulic-Cement, Mortar, and Concrete*
- ASTM C641, *Standard Test Method for Iron Staining Materials in Light-weight Concrete Aggregates*
- ASTM C1262, *Standard Test Method for Evaluating the Freeze-Thaw Durability of Dry-Cast Segmental Retaining Wall Units and Related Concrete Units.*

Concrete mixing and specimen fabrication were conducted in accordance with ASTM C192, *Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory*.

MATERIALS

The as-received lightweight aggregate samples were blended to achieve a combined total aggregate gradation specified in Table X1.1 of ASTM C331. The blended aggregate herein referred to as 'C331 Blend' was used for chemical composition and physical property tests, and for fabrication of concrete specimens for evaluation of drying shrinkage and freeze-thaw resistance. The gradation of C331 Blend is presented in Table 1 below.

Table 1. ASTM C136 gradation of C331 Blend

Sieve No.	% Retained on Individual Sieve	Cumulative % Retained on Individual Sieve	Cumulative % Passing on Individual Sieve	ASTM C331 Limit, % Retained on Individual Sieve
3/8"	0.0	0.0	100.0	0 – 2
#4	9.8	9.8	90.2	0 – 10
#8	32.2	41.9	58.1	15 – 35
#16	17.2	59.1	40.9	15 – 35
#30	14.4	73.5	26.5	5 – 20
#50	7.4	80.9	19.1	5 – 15
#100	7.4	88.2	11.8	5 – 15
Pan	11.8	100.0	0.0	8 – 20

For concrete batching per section 9.6.1 of ASTM C331, a CTLGroup lab stocked Type I/II portland cement, a lab stocked air entraining admixture, and a lab stocked super plasticizer were used in addition to C331 Blend aggregate.

AGGREGATE TESTING

The results of aggregate specific tests are presented in Table 2.

Table 2. Summary of C331 Blend Aggregate Test Results

Test	Description	Result	ASTM C331 Limit	Requirement Met?
ASTM C40	Organic Impurities, liquid color	Lighter than No. 1	No.1 Maximum	Yes
ASTM C641	Staining Stain Index	0	60 Maximum	Yes
ASTM C142	Clay Lumps and Friable Particles	0.4	2% Maximum	Yes
ASTM C29	Loose Dry Bulk Density, lb/ft^3	50	70 lb/ft^3 Maximum	Yes

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CONCRETE MIXING

The required quantity of C331 Blend aggregate was soaked in water for 72 hours prior to mixing to achieve a saturated condition. Following this, the additional water was allowed to drain from the aggregate container over a period of 24 hours. The wet aggregate was then thoroughly blended, and a test sample was procured and tested to determine the total moisture content. The purpose is to determine the actual water content required for batching through appropriate moisture corrections. The total amount of water was adjusted during mixing to achieve the desired slump as required by ASTM C331 standard. The sequence of concrete mixing was as follows:

- Initial 3 minutes of mixing: lightweight aggregate, cementitious materials, and water (some water was retained for the last 2 minutes of mixing)
- 3 minutes of rest
- 2 minutes of mixing: water was added as required for the specified slump

Soon after fresh properties of concrete (slump, plastic air content, density, and temperature), three 2x2x11.25-in. and five 3x3x11.25-in. nominal prisms were fabricated for drying shrinkage and freeze-thaw tests, respectively. The quantities of different materials used to produce concrete mixture are presented in Table 3.

Table 3. Concrete Mixture Proportions (SSD Condition)

Ingredient	Quantity
Cement, lb/yd^3	733
C331 Blend, lb/yd^3	1626
Water, lb/yd^3	298
Resulting w/cm ratio	0.40

FRESH CONCRETE PROPERTIES

After mixing, fresh concrete properties were measured as follows:

- Slump according to ASTM C143, *Standard Test Method for Slump of Hydraulic Cement Concrete*
- Unit weight (density) according to ASTM C138, *Standard Test Method for Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete*

The measured fresh concrete properties are presented in Table 4.

Table 4. Fresh Concrete Properties

Property	Test Value
Measured Slump, in.	2.00
Unit Weight (Density), lb/ft^3	97.3
Gravimetric Air, %	6.0

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HARDENED CONCRETE PROPERTIES

A summary of the tests results conducted on hardened concrete are presented in Table 5.

Table 5. Hardened Concrete Properties

Property	Test Result	ASTM C331 Limit	Requirement met?
ASTM C157 Shrinkage, %	0.09%	0.10%	Yes
ASTM C151 Autoclave Expansion, %	Test Not Performed	No pop outs	—
ASTM C1262 Freeze-thaw Resistance, Cumulative Weight Loss, %	0.8%	—	—

CONCLUSION

The results of the testing program indicate **the C331 Blend aggregate, which represents a blend of different size fractions of 'eco Brick Innovio' aggregate supplied by Eco Brick, met the performance limit requirements specified by ASTM C331. The testing scope excluded Loss on Ignition (ASTM C140) and Autoclave Expansion (ASTM C151) procedures.**

Concrete specimens produced using C331 Blend aggregate and proportioned per ASTM C331 were evaluated for freeze-thaw resistance (ASTM C1262), and drying shrinkage (ASTM C157). For ASTM C1262 testing, the measured average cumulative mass loss after 100 freeze-thaw cycles was 93, indicating that all test specimens exhibited adequate or satisfactory freeze thaw performance. The average drying shrinkage recorded at 100 days of age was 0.09%, which was below the specified ASTM C331 limit (0.10%).

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CLOSING

Thank you for having CTLGroup assist you on this project. Should you have any questions, please call or email.

Sincerely,
CTLGroup



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Attachments: Test reports (11 pages)

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